

# Flatness-Based and Passivity-Based Control with Adaptive Neural Compensation of a Direct-Drive PMSG Wind Energy Conversion System

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**Abstract**—This paper proposes a control strategy for a grid-connected 2 MW direct-drive permanent magnet synchronous generator (PMSG) wind turbine with a back-to-back converter and an  $L$  filter. The outer loops are designed by differential flatness: the rotor speed is the flat output of the maximum power point tracking (MPPT) loop, and the energy stored in the DC link and in the filter is the flat output of the DC-link loop. The inner current loops of both converters use passivity-based control (PBC), which keeps the workless coupling of the machine and filter models and injects damping; a storage function proves their stability. An adaptive 6-10-2 neural network, updated online with dead-zone,  $\sigma$ -modification and back-propagation, adds a compensation signal to the flat loops without changing the nominal gains. Simulations under turbulent wind, parameter mismatch, a 10 % error on the aerodynamic model and a 20 % grid voltage dip are compared with PI vector control. The proposed control tracks the MPPT speed 3.8 times more accurately and limits the DC-link deviation during the dip to 3.6 V, against 11.6 V with PI and power feedforward, while the neural compensation reduces the integral absolute speed error by 38 %.

**Index Terms**—PMSG, Wind Energy Conversion Systems, Back-to-Back Converter, Flatness-Based Control, Passivity-Based Control, Adaptive Neural Network, Maximum Power Point Tracking (MPPT).

## I. INTRODUCTION

Wind energy is one of the main sources in the transition towards low-carbon electricity generation [1]; a record 117 GW of new wind capacity was installed worldwide in 2023 [2]. Direct-drive PMSG turbines remove the gearbox, which reduces maintenance and losses, and their full-scale back-to-back converter decouples the generator from the grid [3], [4]. This converter must extract the maximum power from the wind, regulate the DC-link voltage and inject a clean current into the grid.

These tasks are usually performed by cascaded PI loops in rotating frames [5], [6]. The PMSG model is nonlinear and coupled, and its parameters vary with temperature and saturation. Nonlinear controllers have therefore been studied for wind turbines and PMSGs, such as backstepping [7], predictive control [8] and passivity-based control [9].

Differential flatness [10] computes the control input directly from a desired trajectory of a flat output; using the stored

energy as the flat output gives a simple linear power balance for DC buses [11]. Passivity-based control [12], [13] shapes the energy of the closed loop and keeps the physical structure of the system. Both methods rely on a model, so model errors leave residual tracking errors. Adaptive neural networks can learn such residual terms online [14]; in [15], an adaptive neural compensation was added to a nominal controller of a DFIG without changing its gains.

This paper combines these tools for the complete chain of a 2 MW direct-drive PMSG turbine. The individual techniques are known; the contributions are their adaptation to this chain and the resulting analysis:

- a DC-link flat output that includes the energy of the  $L$  filter, so that the grid-side law contains the measured generator power as an exact feedforward and the DC-link error dynamics is linear;
- PBC laws for the currents of both converters and a cascade proof that the complete nominal interconnection (flat outer loops and PBC inner loops) is exponentially stable, which is the assumption required by the neural compensation of [15];
- the transfer of this compensation, developed in [15] for the rotor currents of a DFIG, to the two flat loops of a PMSG, with a learning signal taken along their control directions. The network structure and its update laws are those of [15];
- an evaluation against PI vector control with and without power feedforward, over several wind realizations, initial conditions and adaptation settings, and with a switched converter model.

The model is given in Sections II–III, the wind profile and the references in Section IV, the control and its stability in Section V, and the simulations in Section VI.

## II. WIND TURBINE MODELING

The aerodynamic power captured by the turbine is [17]

$$P_w = \frac{1}{2} \rho \pi R^2 C_p(\lambda, \beta) v^3 \quad (1)$$

where  $\rho$  is the air density,  $R$  the rotor radius,  $v$  the wind speed,  $\beta$  the pitch angle and  $C_p$  the power coefficient, which depends on the tip-speed ratio

$$\lambda = \frac{\Omega R}{v} \quad (2)$$

with  $\Omega$  the rotor speed. The power coefficient is modeled by

$$C_p = 0.5176 \left( \frac{116}{\lambda_i} - 0.4\beta - 5 \right) e^{-21/\lambda_i} + 0.0068\lambda \quad (3)$$

with  $1/\lambda_i = 1/(\lambda + 0.08\beta) - 0.035/(\beta^3 + 1)$ . For  $\beta = 0$ , its maximum  $C_{p,\max} = 0.48$  is reached at  $\lambda_{opt} = 8.1$ . The aerodynamic torque is  $T_w = P_w/\Omega$ , and the direct-drive shaft is described by the one-mass model

$$J \frac{d\Omega}{dt} = T_w - T_e - B\Omega \quad (4)$$

where  $J$  is the total inertia,  $T_e$  the electromagnetic torque and  $B$  the viscous friction coefficient. With  $R = 40$  m and  $\Omega_n = 2.23$  rad/s, the rated wind speed is  $v_n = \Omega_n R / \lambda_{opt} \approx 11$  m/s and the captured power at  $v_n$  is about 2 MW.

### III. PMSG AND BACK-TO-BACK CONVERTER MODEL

#### A. PMSG

In the rotor  $dq$  frame, with the  $d$  axis aligned with the magnet flux, generator convention and  $L_d = L_q = L_s$ , the PMSG is described by [4]

$$L_s \frac{di_{sd}}{dt} = -R_s i_{sd} + \omega_e L_s i_{sq} - v_{sd} \quad (5)$$

$$L_s \frac{di_{sq}}{dt} = -R_s i_{sq} - \omega_e L_s i_{sd} + \omega_e \psi_f - v_{sq} \quad (6)$$

where  $\omega_e = p\Omega$ ,  $p$  is the number of pole pairs and  $\psi_f$  the magnet flux. The electromagnetic torque is

$$T_e = \frac{3}{2} p \psi_f i_{sq}. \quad (7)$$

With  $\mathbf{i}_s = [i_{sd} \ i_{sq}]^T$ ,  $\mathbf{v}_s = [v_{sd} \ v_{sq}]^T$ ,  $\mathbf{e}_q = [0 \ 1]^T$  and the skew-symmetric matrix  $\mathbf{J} = \begin{bmatrix} 0 & 1 \\ -1 & 0 \end{bmatrix}$ , the model reads

$$L_s \dot{\mathbf{i}}_s = -R_s \mathbf{i}_s + \omega_e L_s \mathbf{J} \mathbf{i}_s + \omega_e \psi_f \mathbf{e}_q - \mathbf{v}_s. \quad (8)$$

Since  $\mathbf{i}_s^T \mathbf{J} \mathbf{i}_s = 0$ , the coupling term  $\omega_e L_s \mathbf{J} \mathbf{i}_s$  is workless: it exchanges energy between the two axes without producing or dissipating it.

#### B. Back-to-Back Converter, DC Link and Grid

The two-level converters are represented by their average model. The voltages they synthesize are limited by the space-vector modulation range,  $\|\mathbf{v}_s\|, \|\mathbf{v}_i\| \leq V_{dc}/\sqrt{3}$ , and, neglecting losses, the DC-link capacitor obeys

$$CV_{dc} \frac{dV_{dc}}{dt} = P_{msc} - P_{gsc} \quad (9)$$

with  $P_{msc} = \frac{3}{2} \mathbf{v}_s^T \mathbf{i}_s$  and  $P_{gsc} = \frac{3}{2} \mathbf{v}_i^T \mathbf{i}_g$ . In the grid-voltage-oriented frame ( $v_{gd} = V_m$ ,  $v_{gq} = 0$ ), the  $L$  filter gives

$$L_f \dot{\mathbf{i}}_g = -R_f \mathbf{i}_g + \omega_g L_f \mathbf{J} \mathbf{i}_g + \mathbf{v}_i - \mathbf{v}_g \quad (10)$$

and the powers injected into the grid are  $P = \frac{3}{2} v_{gd} i_{gd}$  and  $Q = -\frac{3}{2} v_{gd} i_{gq}$ .

TABLE I  
PARAMETERS OF THE STUDIED SYSTEM

| Parameter                   | Value                               | Parameter  | Value                 |
|-----------------------------|-------------------------------------|------------|-----------------------|
| $P_n$                       | 2 MW                                | $p$        | 26                    |
| $R$                         | 40 m                                | $R_s$      | 0.78 m $\Omega$       |
| $\Omega_n$                  | 2.23 rad/s                          | $L_s$      | 1.57 mH               |
| $T_n$                       | 0.897 MN·m                          | $\psi_f$   | 9.18 Wb               |
| $\lambda_{opt}, C_{p,\max}$ | 8.1, 0.48                           | $C$        | 20 mF                 |
| $\rho$                      | 1.225 kg/m <sup>3</sup>             | $V_{dc}^*$ | 1200 V                |
| $J$                         | $2.8 \times 10^6$ kg·m <sup>2</sup> | $L_f, R_f$ | 0.15 mH, 2 m $\Omega$ |
| $B$                         | $10^3$ N·m·s/rad                    | Grid       | 690 V, 50 Hz          |

#### C. Complete Model

With the state  $\mathbf{x} = [i_{sd} \ i_{sq} \ \Omega \ V_{dc} \ i_{gd} \ i_{gq}]^T$ , the inputs  $\mathbf{u} = [\mathbf{v}_s^T \ \mathbf{v}_i^T]^T$  and the disturbances  $v$  and  $\mathbf{v}_g$ , (4), (8), (9) and (10) form the sixth-order nonlinear model

$$\begin{cases} \dot{i}_{sd} = \frac{1}{L_s} (-R_s i_{sd} + p\Omega L_s i_{sq} - v_{sd}) \\ \dot{i}_{sq} = \frac{1}{L_s} (-R_s i_{sq} - p\Omega L_s i_{sd} + p\Omega \psi_f - v_{sq}) \\ \dot{\Omega} = \frac{1}{J} (T_w(\Omega, v) - \frac{3}{2} p \psi_f i_{sq} - B\Omega) \\ \dot{V}_{dc} = \frac{1}{CV_{dc}} (P_{msc} - P_{gsc}) \\ \dot{i}_{gd} = \frac{1}{L_f} (-R_f i_{gd} + \omega_g L_f i_{gq} + v_{id} - v_{gd}) \\ \dot{i}_{gq} = \frac{1}{L_f} (-R_f i_{gq} - \omega_g L_f i_{gd} + v_{iq} - v_{gq}) \end{cases} \quad (11)$$

The parameters are listed in Table I. The turbine data correspond to a standard 2 MW turbine; the generator, DC-link, filter and grid data are those of the non-salient PMSG of [16]. The magnet flux is consistent with the grid: at 9.75 Hz, the rated frequency of [16], the back-EMF  $\psi_f \omega_e = 562$  V equals the peak phase voltage of the 690 V grid.

#### IV. WIND PROFILE AND REFERENCE GENERATION

The wind speed combines a mean value, low-frequency harmonics and a turbulence term [15]:

$$v(t) = V_{mean} + \sum_{k=1}^3 A_k \sin(2\pi f_k t) + v_{turb}(t) \quad (12)$$

with  $V_{mean} = 8$  m/s,  $A = [1.5, 0.2, 0.3]$  m/s,  $f = [0.03, 0.16, 0.17]$  Hz, and  $v_{turb}$  a smoothed Gaussian noise (standard deviation 0.01 m/s, time constant 0.05 s). The wind rises from 8 m/s to 9 m/s at  $t \approx 2$  s, falls to 8.6 m/s at  $t \approx 4.2$  s and reaches 10 m/s at  $t \approx 7.7$  s.

The maximum power is extracted at  $\lambda = \lambda_{opt}$ . The measured wind is filtered,  $\tau_v \dot{\hat{v}} = v - \hat{v}$  with  $\tau_v = 0.3$  s, and the speed reference is obtained from the critically damped filter

$$\ddot{\Omega}^* + 2\omega_r \dot{\Omega}^* + \omega_r^2 \Omega^* = \omega_r^2 \frac{\lambda_{opt} \hat{v}}{R} \quad (13)$$

with  $\omega_r = 1.5$  rad/s, which provides  $\Omega^*$  and  $\dot{\Omega}^*$  without numerical differentiation. Since  $T_e$  only depends on  $i_{sq}$ ,  $i_{sd}^* = 0$ ; unity power factor gives  $i_{gq}^* = 0$ ; the DC-link reference is  $V_{dc}^* = 1200$  V.

#### V. PROPOSED CONTROL

The control structure is shown in Fig. 1. It is sampled at  $T_s = 100 \mu\text{s}$ .

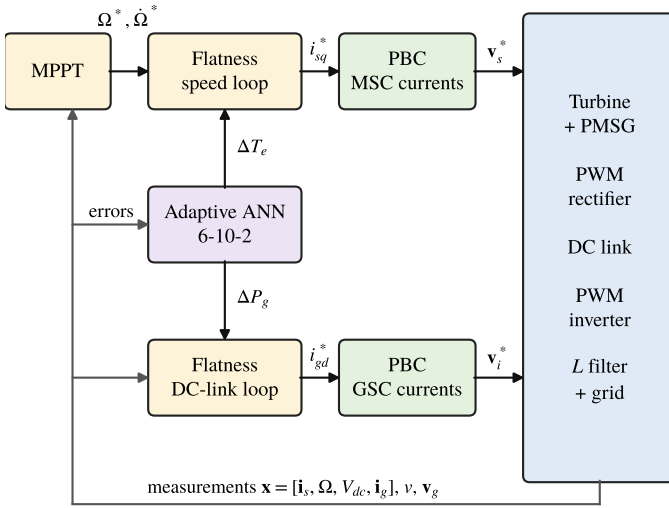


Fig. 1. Proposed control: flat outer loops, PBC current loops and adaptive neural compensation.

### A. Flatness-Based Outer Loops

A system is flat if its states and inputs can be written as functions of a flat output and of a finite number of its derivatives [10]. The control is then obtained by inverting the model with a desired dynamics imposed on the flat output.

*Speed loop.* From (4),  $T_e = T_w - B\Omega - J\dot{\Omega}$ , so  $y_1 = \Omega$  is a flat output. Replacing  $\dot{\Omega}$  by

$$\nu_1 = \dot{\Omega}^* + k_{11}e_\Omega + k_{12} \int e_\Omega dt, \quad e_\Omega = \Omega^* - \Omega \quad (14)$$

gives

$$T_e^* = \hat{T}_w - B\Omega - J\nu_1 + \Delta T_e, \quad i_{sq}^* = \frac{T_e^*}{\frac{3}{2}p\psi_f} \quad (15)$$

where  $\hat{T}_w$  is computed from (1)–(3) with the measured wind and  $\Delta T_e$  is the neural compensation.  $T_e^*$  is limited to  $[0, 1.2T_n]$  and the integral is frozen during saturation. If  $\hat{T}_w = T_w$  and  $\Delta T_e = 0$ , the error obeys  $\dot{e}_\Omega + k_{11}e_\Omega + k_{12} \int e_\Omega dt = 0$ .

*DC-link loop.* The flat output is the energy stored in the DC link and in the filter,

$$y_2 = \frac{1}{2}CV_{dc}^2 + \frac{3}{4}L_f\|\mathbf{i}_g\|^2. \quad (16)$$

From (9) and (10), using  $\mathbf{i}_g^T \mathbf{J} \mathbf{i}_g = 0$ ,

$$\dot{y}_2 = P_{msc} - \frac{3}{2}v_{gd}i_{gd} - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 \quad (17)$$

which is linear in the injected power, as in [11]. With  $e_y = y_2^* - y_2$  and  $\dot{y}_2^* \approx 0$ , imposing  $\dot{e}_y = -\nu_2$ , i.e.  $\dot{y}_2 = \nu_2$ , with  $\nu_2 = k_{21}e_y + k_{22} \int e_y dt$ , yields

$$P_g^* = P_{msc} - \frac{3}{2}R_f\|\mathbf{i}_g\|^2 - \nu_2 + \Delta P_g, \quad i_{gd}^* = \frac{2P_g^*}{3v_{gd}}. \quad (18)$$

Inserting (18) into the expression of  $\dot{y}_2$  with  $i_{gd} = i_{gd}^*$  gives  $\dot{y}_2 = \nu_2 - \Delta P_g$ , hence  $\dot{e}_y + k_{21}e_y + k_{22} \int e_y dt = \Delta P_g$ . The term  $P_{msc}$ , known from the applied voltages and measured

currents, anticipates the power coming from the generator. The reference  $y_2^* = \frac{1}{2}CV_{dc}^{*2} + \frac{3}{4}L_f\|\mathbf{i}_g^*\|^2$  varies only through the small filter energy, whose derivative is neglected. The gains are  $k_{11} = 2\zeta_\Omega\omega_\Omega$ ,  $k_{12} = \omega_\Omega^2$ ,  $k_{21} = 2\zeta_v\omega_v$ ,  $k_{22} = \omega_v^2$  (Table II).

### B. Passivity-Based Current Loops

For the machine-side converter, the error  $\mathbf{e}_s = \mathbf{i}_s - \mathbf{i}_s^*$  and its integral  $\mathbf{z}_s$  ( $\dot{\mathbf{z}}_s = \mathbf{e}_s$ ) are used in

$$\mathbf{v}_s^* = -L_s\dot{\mathbf{i}}_s^* - R_s\mathbf{i}_s^* + \omega_e L_s \mathbf{J} \mathbf{i}_s^* + \omega_e \psi_f \mathbf{e}_q + R_a \mathbf{e}_s + K_i \mathbf{z}_s \quad (19)$$

where  $R_a$  is the injected damping. For the grid-side converter, with  $\mathbf{e}_g = \mathbf{i}_g - \mathbf{i}_g^*$  and  $\dot{\mathbf{z}}_g = \mathbf{e}_g$ ,

$$\mathbf{v}_i^* = L_f\dot{\mathbf{i}}_g^* + R_f\mathbf{i}_g^* - \omega_g L_f \mathbf{J} \mathbf{i}_g^* + \mathbf{v}_g - R_b \mathbf{e}_g - K_{ig} \mathbf{z}_g. \quad (20)$$

The reference derivatives are obtained by a first-order filter ( $\tau_d = 2$  ms) and the voltages are limited to  $V_{dc}/\sqrt{3}$ , with the integrals frozen at the limit.

**Proposition 1.** With (19) applied to (8) and  $R_a, K_i > 0$ , the storage function  $H = \frac{1}{2}L_s\|\mathbf{e}_s\|^2 + \frac{1}{2}K_i\|\mathbf{z}_s\|^2$  satisfies  $\dot{H} = -(R_s + R_a)\|\mathbf{e}_s\|^2 \leq 0$ , and  $\mathbf{e}_s \rightarrow 0$ . The same holds for (20) applied to (10).

*Proof.* Subtracting  $L_s\dot{\mathbf{i}}_s^*$  from (8) and inserting (19) gives  $L_s\dot{\mathbf{e}}_s = -(R_s + R_a)\mathbf{e}_s + \omega_e L_s \mathbf{J} \mathbf{e}_s - K_i \mathbf{z}_s$ . Hence  $\dot{H} = \mathbf{e}_s^T L_s \dot{\mathbf{e}}_s + K_i \mathbf{z}_s^T \dot{\mathbf{e}}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2 + \omega_e L_s \mathbf{e}_s^T \mathbf{J} \mathbf{e}_s$ , and the last term vanishes because  $\mathbf{J}$  is skew-symmetric.  $H$  is bounded and  $\dot{H}$  is uniformly continuous, so Barbalat's lemma [18] gives  $\mathbf{e}_s \rightarrow 0$ .  $\square$

### C. Adaptive Neural Compensation

As in [15], the network does not replace the nominal control  $u_N$  given by (15)–(18) and does not modify its gains; it adds

$$u = u_N + u_{AI}, \quad u_{AI} = [\Delta T_e \ \Delta P_g]^T. \quad (21)$$

Its six inputs are built from the errors  $\bar{\mathbf{e}} = [e_\Omega/(0.01\Omega_n), e_y/(300 \text{ J}), e_{isq}/(0.01I_n)]^T$ , normalized by 1% of the rated speed, 300 J (about 1% of  $V_{dc}^*$ ) and 1% of the rated current  $I_n = T_n/(\frac{3}{2}p\psi_f)$ , and from their derivatives, filtered with a 5 ms time constant and scaled by  $\tau = [0.5, 0.02, 0.002]$  s:  $\mathbf{z}_n = [\bar{\mathbf{e}}^T, (\tau \odot \dot{\bar{\mathbf{e}}})^T]^T$ . With one hidden layer of ten neurons,

$$\mathbf{h} = \tanh(W_1 \mathbf{z}_n + \mathbf{b}_1), \quad \mathbf{y} = W_2 \mathbf{h} + \mathbf{b}_2, \quad (22)$$

$$u_{AI} = K_{AI} \text{diag}(0.1T_n, 0.1P_n) \text{sat}_{\pm 3}(\mathbf{y}) \quad (23)$$

with  $W_1 \in \mathbb{R}^{10 \times 6}$  and  $W_2 \in \mathbb{R}^{2 \times 10}$ .  $T_e$  and  $P_g$  enter  $\dot{e}_\Omega$  and  $\dot{e}_y$  with a positive sign, so the learning signal in the control direction is  $\mathbf{r} = [\bar{e}_1 \ \bar{e}_2]^T$ , set to zero inside a dead zone  $\|\mathbf{r}\| \leq \delta$ . The parameters are updated online by

$$\dot{W}_2 = -\Gamma_n \mathbf{r} \mathbf{h}^T - \sigma W_2, \quad \dot{\mathbf{b}}_2 = -\Gamma_n \mathbf{r} - \sigma \mathbf{b}_2, \quad (24)$$

$$\dot{W}_1 = -\Gamma_n \delta_h \mathbf{z}_n^T - \sigma W_1, \quad \dot{\mathbf{b}}_1 = -\Gamma_n \delta_h - \sigma \mathbf{b}_1, \quad (25)$$

with  $\delta_h = (1 - \mathbf{h}^2) \odot (W_2^T \mathbf{r})$  and the normalized gain  $\Gamma_n = \Gamma/(1 + \|\mathbf{z}_n\|^2)$ . The  $\sigma$ -modification terms [14], [19] prevent parameter drift, and the normalization [19] slows the adaptation when the errors are large, e.g. after a torque

saturation; with a constant gain, a start 10 % below the MPPT speed led to an oscillatory adaptation in simulation. Compared with [15], this normalization is the only change to the update laws. The adaptation is frozen when  $T_e^*$  or  $i_{gd}^*$  saturates, and  $u_{AI}$  is applied at the next sample.  $W_2$  and  $\mathbf{b}_2$  start at zero, so the compensation is initially inactive.

**Proposition 2.** Consider (11) with (15)–(20),  $\Delta T_e = \Delta P_g = 0$ , an exact model, no saturation, exact reference derivatives and  $|\omega_e| \leq \bar{\omega}$  on the operating region. Then the origin of the error state  $\xi = (\mathbf{e}_s, \mathbf{z}_s, \mathbf{e}_g, \mathbf{z}_g, e_\Omega, \int e_\Omega, e_y, \int e_y)$  is exponentially stable.

*Proof. Inner loops.* With  $V_s = H + \varepsilon \mathbf{e}_s^T L_s \mathbf{z}_s$ ,  $\dot{V}_s = -(R_s + R_a)\|\mathbf{e}_s\|^2 + \varepsilon(L_s\|\mathbf{e}_s\|^2 - (R_s + R_a)\mathbf{e}_s^T \mathbf{z}_s - \omega_e L_s \mathbf{e}_s^T \mathbf{J} \mathbf{z}_s - K_i \|\mathbf{z}_s\|^2)$ . Since  $|\omega_e| \leq \bar{\omega}$ , a small  $\varepsilon > 0$  makes  $V_s$  positive definite and  $\dot{V}_s \leq -c(\|\mathbf{e}_s\|^2 + \|\mathbf{z}_s\|^2)$  with  $c > 0$ , so  $(\mathbf{e}_s, \mathbf{z}_s) \rightarrow 0$  exponentially; the same holds for  $(\mathbf{e}_g, \mathbf{z}_g)$  since  $\omega_g$  is constant. *Outer loops.* With the actual currents,  $T_e = T_e^* + \frac{3}{2}p\psi_f e_{sq}$  and  $i_{gd} = i_{gd}^* + e_{gd}$ , so (4) and the  $y_2$  dynamics give

$$\begin{aligned} \dot{e}_\Omega + k_{11}e_\Omega + k_{12}\int e_\Omega &= \frac{3p\psi_f}{2J}e_{sq}, \\ \dot{e}_y + k_{21}e_y + k_{22}\int e_y &= \frac{3}{2}v_{gd}e_{gd}. \end{aligned}$$

These are exponentially stable linear systems ( $k_{ij} > 0$ ) driven by exponentially decaying inputs with bounded gains. The inner error dynamics do not depend on the outer states, so the interconnection is a cascade of exponentially stable systems and is exponentially stable [18].  $\square$

Proposition 2 provides the assumption of the following result locally, i.e., as long as the limits are not reached. A bounded model error  $\Delta = T_w - \hat{T}_w$  adds the input  $-\Delta/J$  to the speed error equation; by linearity the error then remains bounded, and the integral removes a constant  $\Delta$ .

**Theorem 1** ([15]). Assume that the nominal control makes the tracking error exponentially stable, that the residual to be compensated is approximated by the network with a bounded error on the operating region, and that  $\Gamma, \sigma > 0$ . Then, under the above laws with the dead zone, the tracking error is uniformly ultimately bounded.

**Remark 1.** In simulation,  $\Gamma \geq 20$  destabilizes the DC-link loop; the bound of the theorem requires bounded network parameters, which a moderate adaptation gain ensures.  $\Gamma = 5$  is used.

## VI. NUMERICAL SIMULATIONS

### A. Setup

The model (11) is integrated with a fourth-order Runge–Kutta scheme at step  $T_s$  over 10 s, starting from the MPPT operating point at  $v = 8$  m/s. Two cases are considered:

- *nominal*: the controller uses the true parameters;
- *robustness*: the controller keeps the nominal values while the plant has  $R_s \times 1.5$ ,  $L_s \times 1.2$ ,  $\psi_f \times 0.92$ ,  $R_f \times 1.5$ ,

TABLE II  
CONTROLLER AND ADAPTIVE ANN PARAMETERS

| Parameter                                 | Value                                                                             |
|-------------------------------------------|-----------------------------------------------------------------------------------|
| Speed loop                                | $\omega_\Omega = 2$ rad/s, $\zeta_\Omega = 0.9$ ( $k_{11} = 3.6$ , $k_{12} = 4$ ) |
| DC-link loop                              | $\omega_v = 60$ rad/s, $\zeta_v = 1$ ( $k_{21} = 120$ , $k_{22} = 3600$ )         |
| PBC (MSC)                                 | $R_a = 0.785$ $\Omega$ , $K_i = 98.1$ $\Omega$ /s                                 |
| PBC (GSC)                                 | $R_b = 0.075$ $\Omega$ , $K_{ig} = 9.38$ $\Omega$ /s                              |
| ANN architecture                          | 6–10–2                                                                            |
| $\Gamma$ , $\sigma$ , $\delta$ , $K_{AI}$ | 5, 0.01, 0.1, 1                                                                   |
| Output saturation                         | $\pm 3$ ( $\pm 0.3T_n$ , $\pm 0.3P_n$ )                                           |

TABLE III  
COMPARISON OF THE CONTROLLERS (10 s)

| Metric                                    | PI     | PI+FF  | Proposed<br>no ANN   | Proposed<br>+ ANN    |
|-------------------------------------------|--------|--------|----------------------|----------------------|
| <i>Nominal</i>                            |        |        |                      |                      |
| IAE of $\Omega$ (rad)                     | 0.0384 | 0.0384 | $2.7 \times 10^{-6}$ | $2.7 \times 10^{-6}$ |
| ISE of $V_{dc}$ ( $V^2$ s)                | 26.4   | 0.095  | 0.085                | 0.085                |
| Max $ V_{dc} - V_{dc}^* $ (V)             | 2.87   | 2.46   | 0.36                 | 0.36                 |
| Energy (kWh)                              | 2.612  | 2.612  | 2.625                | 2.625                |
| <i>Robustness</i>                         |        |        |                      |                      |
| IAE of $\Omega$ (rad)                     | 0.0419 | 0.0419 | 0.0109               | 0.0068               |
| RMS error of $i_{sq}$ (A)                 | 0.84   | 0.84   | 3.05                 | 3.07                 |
| RMS error of $i_{gd}$ (A)                 | 2.15   | 5.98   | 6.88                 | 7.29                 |
| ISE of $V_{dc}$ ( $V^2$ s)                | 255    | 2.69   | 1.30                 | 1.25                 |
| Max $ V_{dc} - V_{dc}^* $ , $t < 1$ s (V) | 4.53   | 3.85   | 10.44                | 10.44                |
| Max $ V_{dc} - V_{dc}^* $ , dip (V)       | 62.3   | 11.6   | 3.35                 | 3.60                 |
| Energy (kWh)                              | 2.599  | 2.599  | 2.615                | 2.614                |

$L_f \times 1.25$ ;  $\hat{T}_w$  is 10 % lower than  $T_w$ ; and the grid voltage drops by 20 % between 8 and 8.2 s.

The proposed control is compared with PI vector control (PI speed loop, PI DC-voltage loop and PI current loops with  $dq$  decoupling [5], [6]), tuned on the same poles as the proposed loops, with and without the feedforward  $P_{msc}/(\frac{3}{2}v_{gd})$  in the DC-voltage loop (PI+FF). All controllers use the same MPPT reference (13) and the same limits. The proposed control is evaluated without and with the neural compensation, with the same nominal gains.

### B. Comparison with PI Vector Control

Table III and Figs. 2–3 summarize the results.

- *Speed (MPPT)*. The flat law contains  $J\dot{\Omega}^*$  and  $\hat{T}_w$ , so the speed follows its reference with an IAE 3.8 times (robustness) to  $10^4$  times (nominal) smaller than with PI, whose speed loop reacts only to the error. The extracted energy is 0.5–0.6 % higher.
- *DC link*. The feedforward of  $P_{msc}$  is decisive: PI without it gives an ISE 200 to 300 times larger. During the grid dip, the deviation is 3.6 V with the proposed control, 11.6 V with PI+FF and 62 V with PI. In steady state, PI+FF keeps a smaller residual ripple (0.03 V against 0.36 V, both below 0.03 % of  $V_{dc}^*$ ).
- *Currents*. The PI loops track the currents more accurately (RMS errors of 0.84 A against 3.07 A for  $i_{sq}$  in the robustness case), because the PBC feedforward terms use the nominal parameters while the PI decoupling uses the

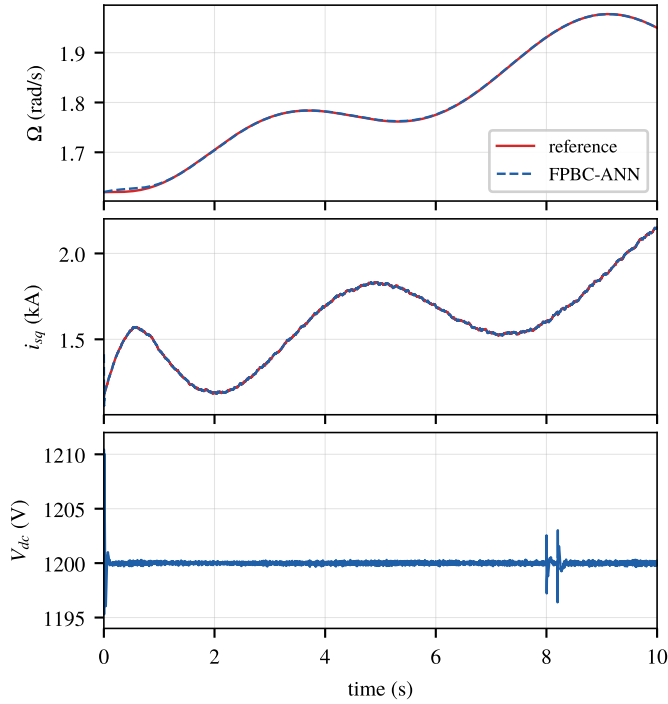


Fig. 2. Robustness case with the proposed control: rotor speed,  $q$ -axis generator current and DC-link voltage.

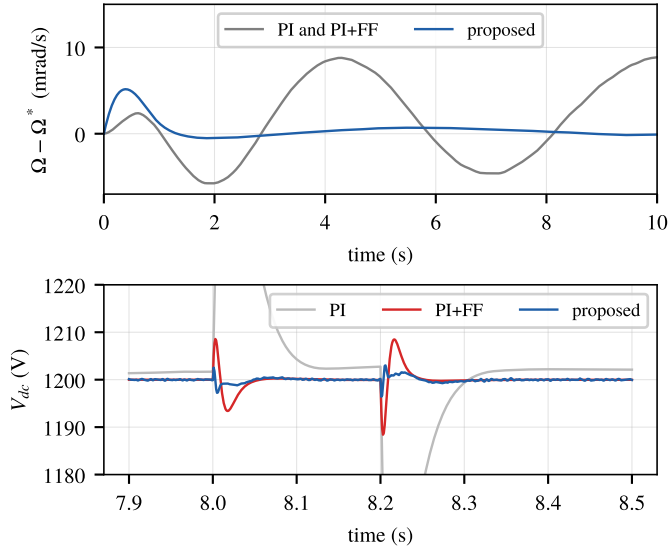


Fig. 3. Robustness case: speed error of the PI controllers and of the proposed control (top), and DC-link voltage during the 20 % grid voltage dip (bottom).

measured currents. All these errors stay below 0.3 % of the rated current.

- *Start-up.* With mismatched parameters, the largest DC-link deviation of the proposed control (10.4 V) occurs during the first instants, against 3.9 V with PI+FF.

### C. Effect of the Neural Compensation

Fig. 4 shows that, without knowing the 10 % model error, the network converges in less than one second to  $\Delta T_e \approx$

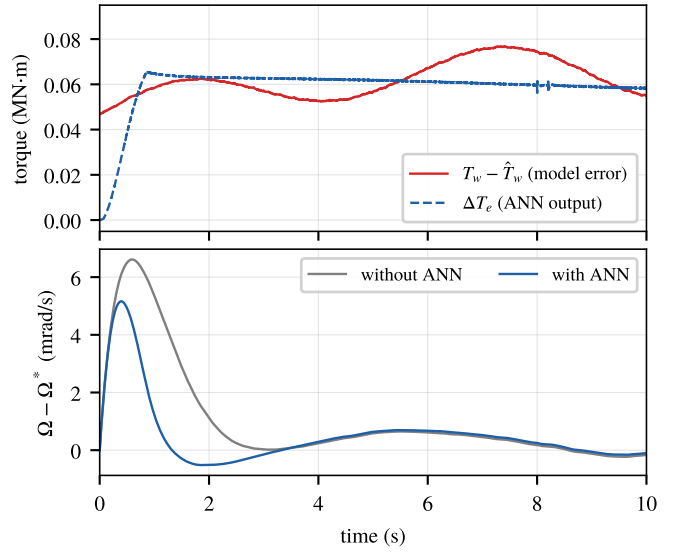


Fig. 4. Robustness case: neural compensation  $\Delta T_e$  compared with the aerodynamic model error, and speed error with and without the network.

TABLE IV  
ADDITIONAL TESTS OF THE NEURAL COMPENSATION (ROBUSTNESS CASE; FIVE WIND REALIZATIONS: MEAN, SPREAD  $< 10^{-5}$  RAD AND  $0.01 \text{ V}^2\text{s}$ ;  $\Omega_0$ : INITIAL SPEED)

| Test                                  | IAE of $\Omega$ (rad)                  | ISE of $V_{dc}$ ( $\text{V}^2\text{s}$ ) | RMS $e_{igd}$ (A) |
|---------------------------------------|----------------------------------------|------------------------------------------|-------------------|
| 5 winds, no ANN                       | 0.01093                                | $1.29 \pm 0.01$                          | –                 |
| 5 winds, ANN                          | 0.00678                                | $1.24 \pm 0.01$                          | –                 |
| $\Gamma = 1$                          | 0.0092                                 | 1.28                                     | 6.96              |
| $\Gamma = 2$                          | 0.0082                                 | 1.27                                     | 7.04              |
| $\Gamma = 5$ (used)                   | 0.0068                                 | 1.25                                     | 7.29              |
| $\Gamma = 10$                         | 0.0061                                 | 1.24                                     | 7.90              |
| $\Gamma = 20$                         | oscillatory ( $V_{dc}$ deviation 47 V) |                                          |                   |
| $\delta = 0.05 / 0.2$                 | 0.0071 / 0.0073                        | 1.23 / 1.25                              | 7.24 / 7.13       |
| $\sigma = 0.001 / 0.1$                | 0.0067 / 0.0083                        | 1.25 / 1.24                              | 7.36 / 7.01       |
| $\Omega_0 = 0.9\Omega^*$ : no ANN/ANN | 0.0921 / 0.0828                        | 1.15 / 1.34                              | 7.95 / 8.02       |
| $\Omega_0 = 0.9\Omega^*$ : PI+FF      | 0.1142                                 | 23.8                                     | 14.2              |

0.065 MN·m, close to the mean of the actual error  $T_w - \hat{T}_w$  (0.047 to 0.077 MN·m); the slowly varying part is handled by the integral action of (15). The IAE of the speed decreases by 38 %, mainly during the start-up transient, where the peak speed error drops from 6.6 to 5.2 mrad/s. Table IV reports additional tests.

Over five turbulence realizations, the reduction of the IAE is  $38 \pm 0.1 \%$  and that of the ISE of  $V_{dc}$  is 4 %. The network acts mainly on the speed loop; it slightly increases the current errors and the dip deviation (Table III), because  $\Delta T_e$  and  $\Delta P_g$  modify the current references. A larger  $\Gamma$  improves the speed tracking but increases the current and DC-link errors, and  $\Gamma = 20$  leads to oscillations:  $\Gamma = 5$  is a compromise. A larger dead zone or a larger  $\sigma$  reduces the compensation. With an initial speed 10 % below the MPPT value, the network reduces the IAE by 10 % and the response remains well damped thanks to the normalized gain. With an initial speed 10 % above, all

controllers saturate the grid-side current at start-up and the DC link rises by more than 120 V, which would require a braking chopper or pitch control.

#### D. Switched Model and Computational Cost

A switched model of the two-level converters was simulated with space-vector modulation, a 2160 Hz carrier, regular sampling with double update and an integration step of about 1  $\mu$ s, at constant wind speeds of 8, 9 and 10 m/s with the nominal parameters. The THD of the grid current, computed by FFT on phase *a* over ten grid periods in steady state up to the 40th harmonic, is 4.4 %, 3.1 % and 2.3 %, respectively, below the 5 % limit of IEEE 519 [20]; PI control gives the same values within 0.04 %, since the ripple is set by the switching frequency and the *L* filter. Per sample, the network requires about 200 multiply–accumulate operations and ten tanh evaluations, and the flat and PBC laws fewer than 100 operations: at 10 kHz, the whole controller needs a few MFLOP/s, within the capability of current digital signal processors.

### VII. CONCLUSION

A complete model of a 2 MW direct-drive PMSG wind energy conversion system, from the turbine to the grid, was developed with machine data from the literature. Its control combines flatness for the speed and DC-link energy loops, passivity-based control for the current loops of both converters, whose stability is proven with a storage function, and an adaptive neural compensation that does not change the nominal gains. Compared with PI vector control tuned on the same poles, the proposed control tracks the MPPT speed 3.8 times more accurately under model errors and limits the DC-link deviation during a 20 % grid dip to 3.6 V (11.6 V with PI and feedforward), at the price of larger current tracking errors. The network learns the aerodynamic torque error online and reduces the speed tracking error by 38 % over several wind realizations, with a normalized gain that keeps the adaptation well damped. Future work will address a braking chopper and pitch control, non-ideal synchronization, a stability analysis including saturations, and experimental validation.

### CONFLICT OF INTEREST

The authors declare there are no conflicts of interest.

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